
Assignment

Probability and Statistics

B.Tech ^{5th} Semester (IT)

1. If the probability density function of a random variable is given by,

$$f(x) = \begin{cases} k(4x - 2x^2), & \text{if } 0 \leq x \leq 2 \\ 0, & \text{elsewhere,} \end{cases}$$

then find the value of k and the probability $P\{X > 1\}$.

2. Discuss conditional probability and illustrate the concept with some detailed examples
3. Under what conditions on variables X and Y , $r(X, Y) = 0$ implies X and Y are independent.
4. A box contains 2^n tickets among which $\binom{n}{i}$ tickets bear the number i , $i = 0, 1, 2, \dots, n$. A group of m tickets is drawn. What is the expectation of sum of their numbers?
5. Buses arrive at a specified stop at 15-minute interval starting at 7 A.M. that is, they arrive at 7, 7:15, 7:30, 7:45 and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits
- (i) less than 5 minutes for a bus;
 - (ii) more than 10 minutes for a bus.
6. Obtain a regression plane by using multiple linear regression to fit the data given below
(independent variable, x: 1 2 3 4)
(independent variable, y: 0 1 2 3)
(dependent variable, z: 12 18 24 30)
7. Discuss normal distribution approximation to binomial distribution and its applications.
8. Define purposive sampling and stratified sampling, illustrate both the concepts with detailed examples. Provide some applications as well.
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